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Lower Limb Nerves

Dr Yasser Al Harbi

Learning Objectives

- Describe the structure and relationships of the plexuses of the lower limb.
- Describe the course, relationships and structures supplied by the major nerves of the lower limb.
- Outline the peripheral nerve lesions, the clinical consequences and giving some examples.

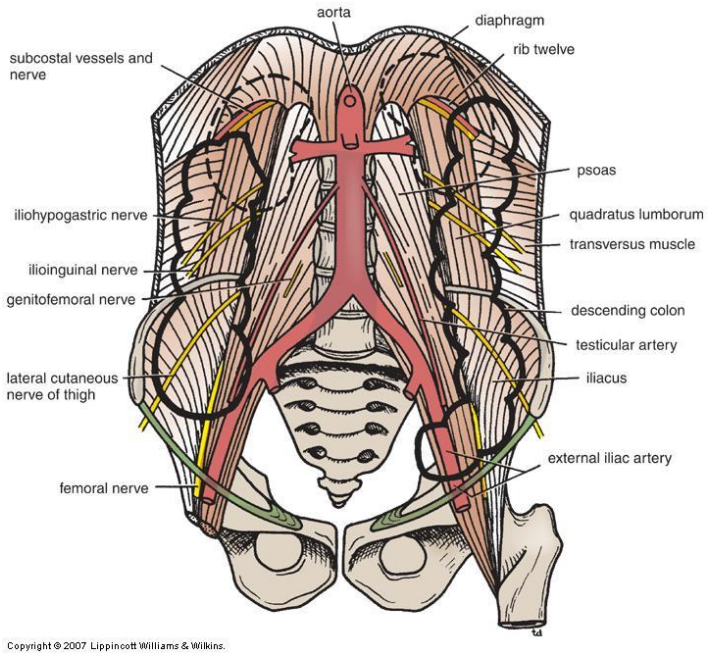
Lumbar Plexus

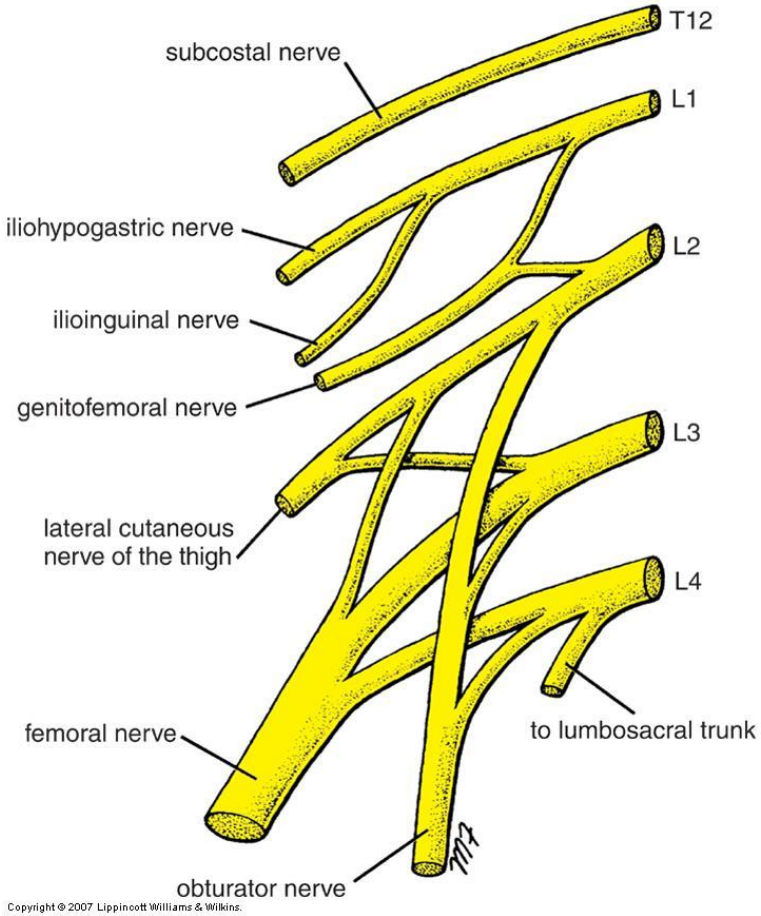
- Ventral rami L1-L4
- Supplies abdominal wall, external genitalia & anterior/medial thigh
- Its branches emerge from around psoas m. :

➢Obturator n. & lumbosacral trunk -----medial border

➢Genitofemoral n. ---- anterior surface

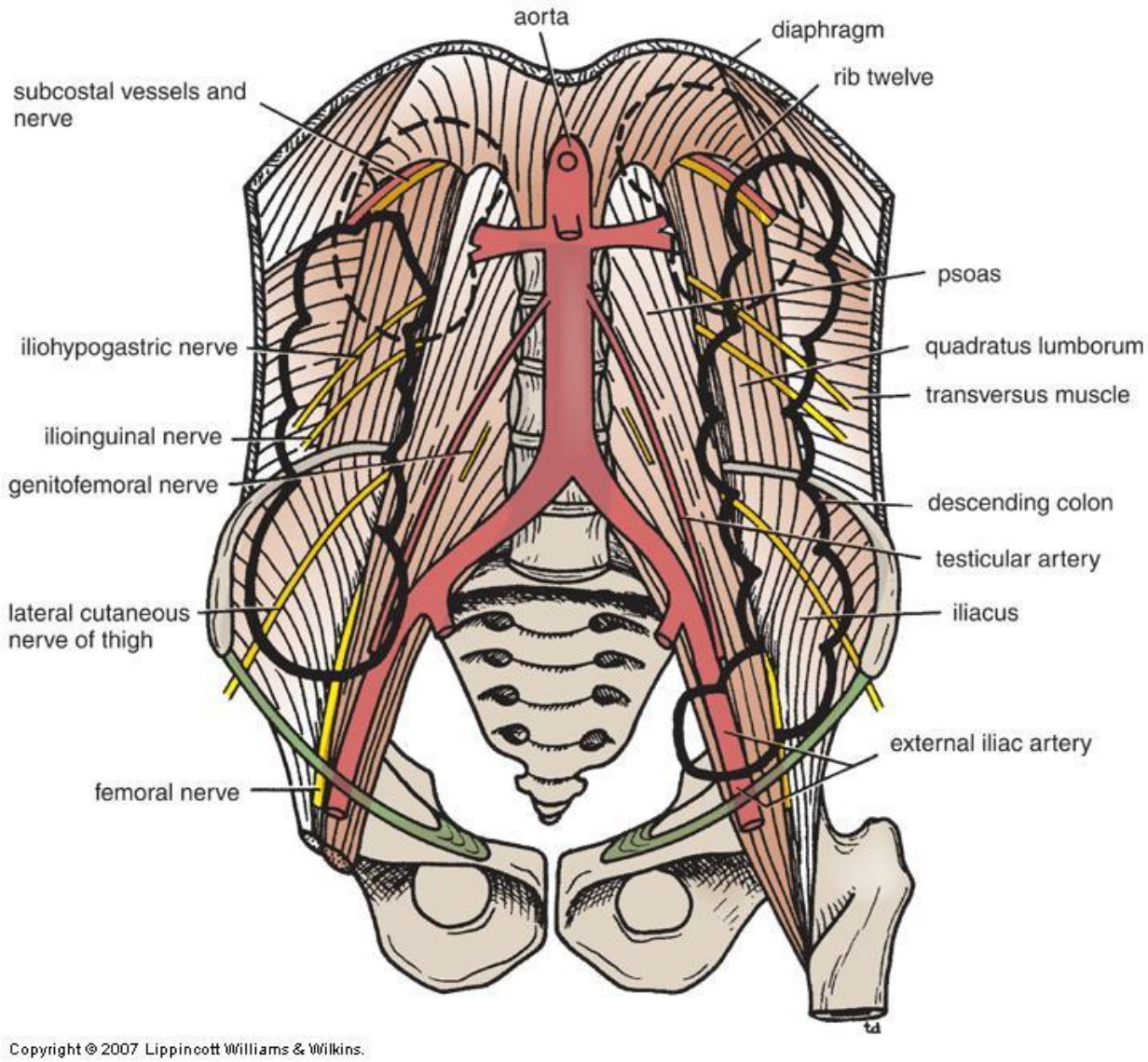
➢Remaining nn. ----- lateral border





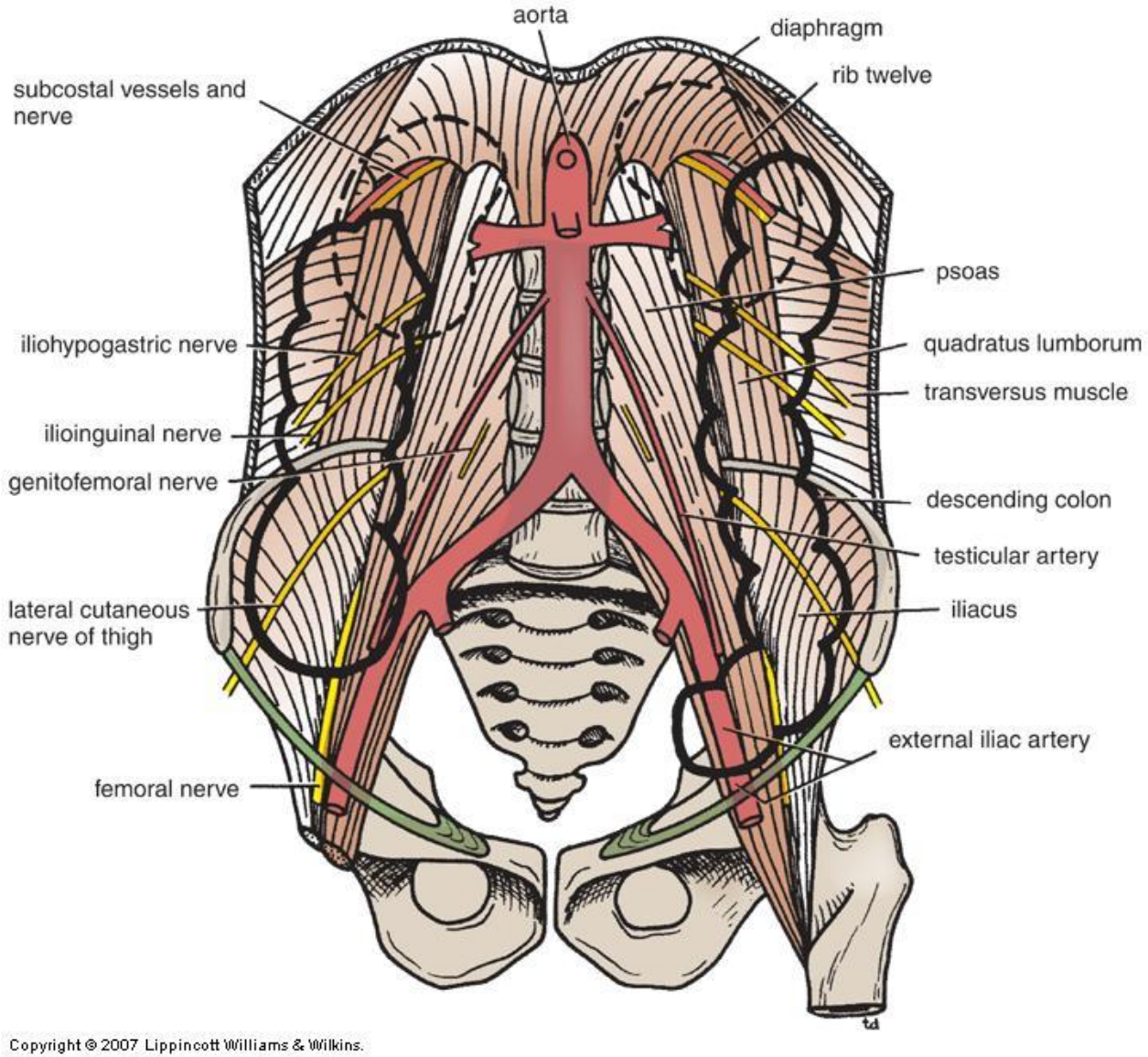
Lumbar Plexus

- Iliohypogastric nerve
  - Relations
    - Psoas m.
    - Quadratus lumborum m.
  - Distribution
    - Muscles of the lateral abdominal wall
    - Skin above the inguinal ligament
- Ilioinguinal nerve
  - Relations
    - Psoas m.
    - Quadratus lumborum m.
    - Inters the inguinal canal and exits from the superficial inguinal ring
  - Distribution
    - Muscles of the lateral abdominal wall
    - Skin above pubic symphysis and scrotum or labia majora



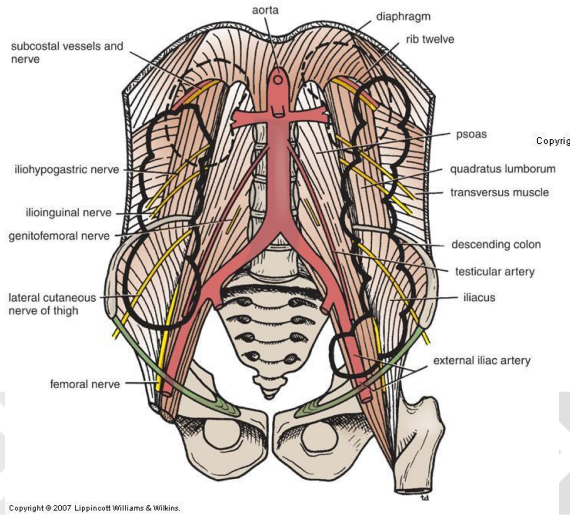
Lumbar Plexus

- Lateral cutaneous nerve of the thigh
  - Relations
    - Psoas m.
    - Iliacus
    - Inguinal ligament
  - Divisions
    - Anterior
    - Posterior
  - Supply the skin over the lateral side of the thigh & knee & lower part of the buttock
- Genitofemoral nerve
  - Relations
    - Psoas m.
  - Divisions
    - Genital branch
      - Enter spermatic cord
      - Supplies cremaster m.
    - Femoral branch
      - Supply small area of skin over the anteriosuperior part of thigh

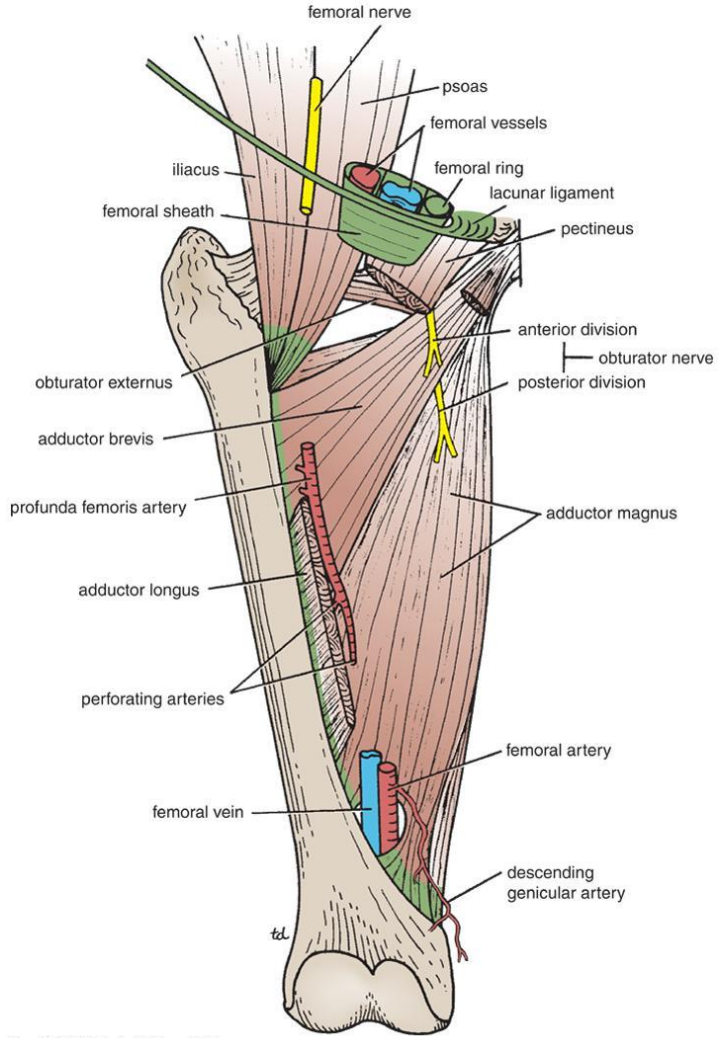


Femoral Nerve

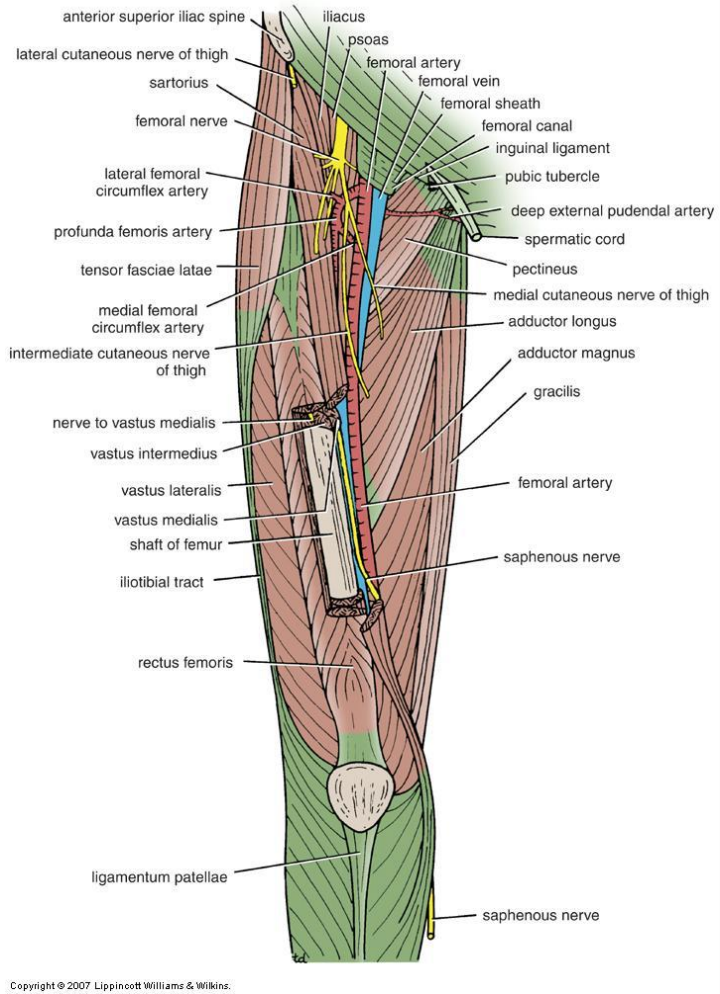
- Largest branch of the lumbar plexus
- Relations
  - Psoas m.
  - Iliacus m.
  - Inguinal ligament
  - Femoral sheath



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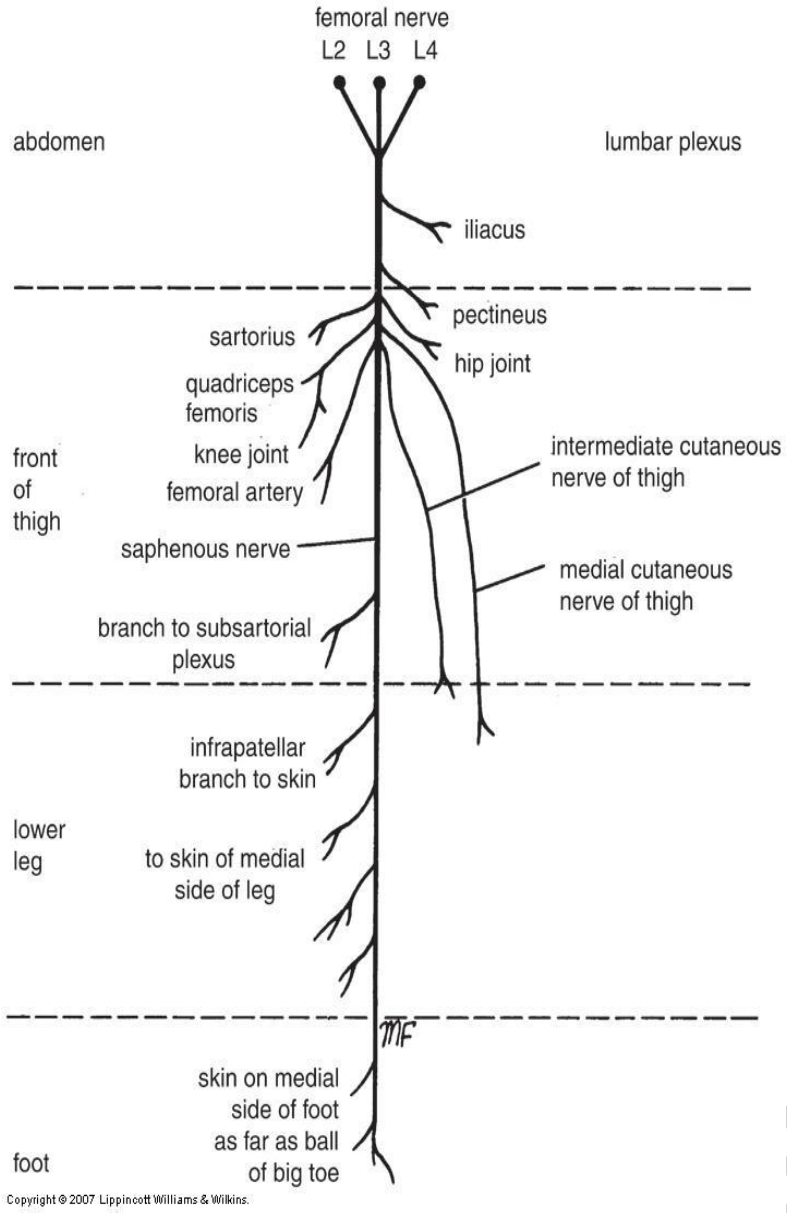
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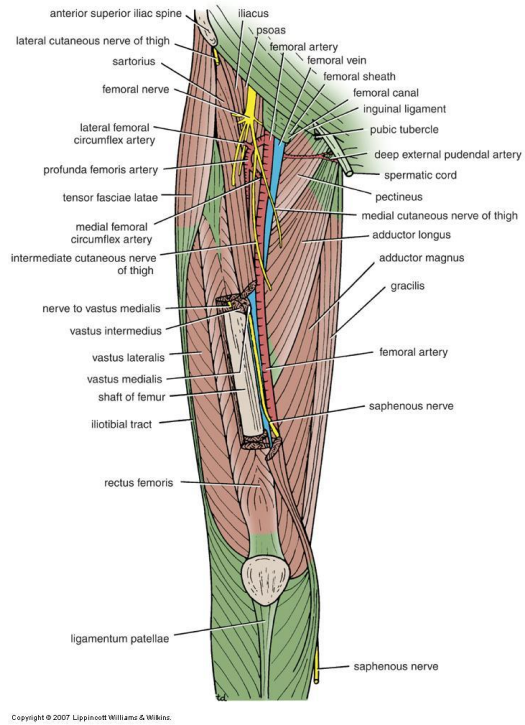


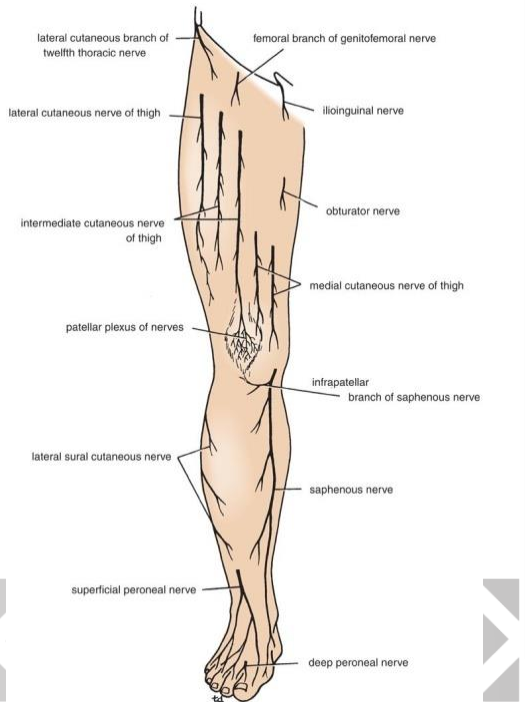
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Femoral Nerve: Branches

- In the abdomen
  - Muscular branch to iliacus
- In the thigh
  - Muscular branches
    - Sartorius, pectineus & quadriceps femoris
  - Cutaneous branches
    - Medial cutaneous n. of the thigh
    - Intermediate cutaneous n. of the thigh
    - Saphenous n.
      - Relations
        - In Femoral triangle
        - Within Adductor canal
        - Cross Femoral a.
        - Between Sartorius & gracilis tendons
        - Companies great saphenous v.
        - Anterior to Medial malleolus
  - Articular branches (hip & knee)

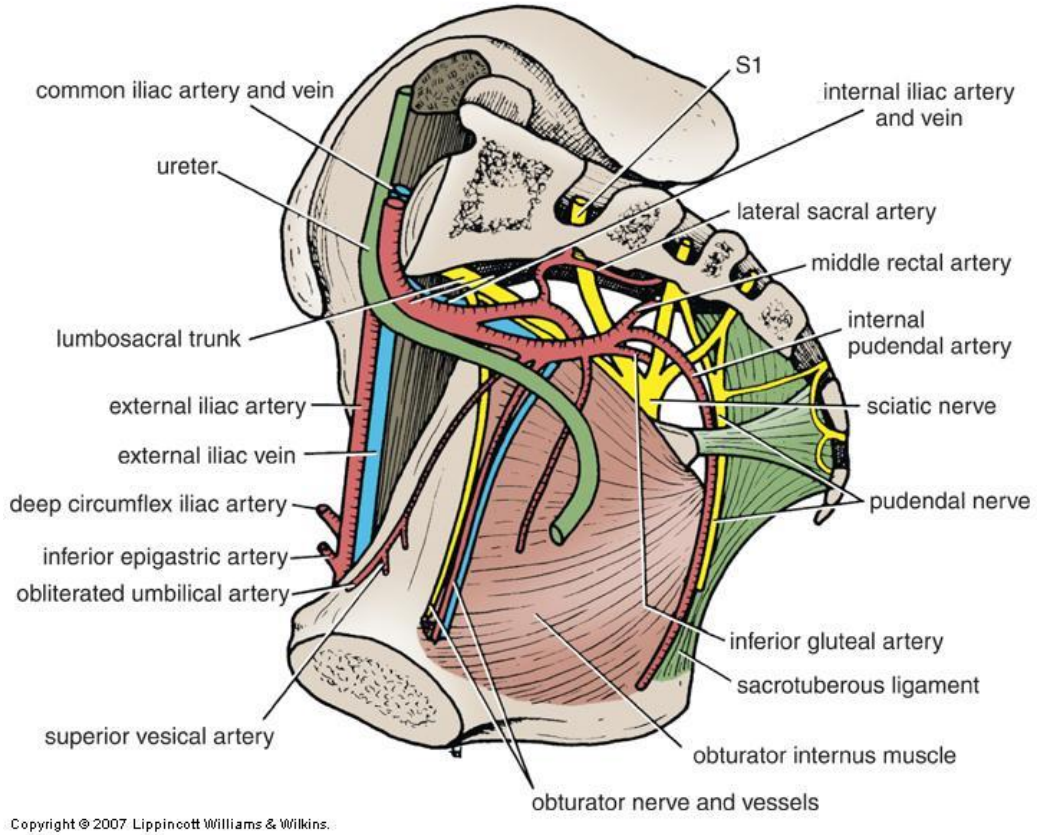




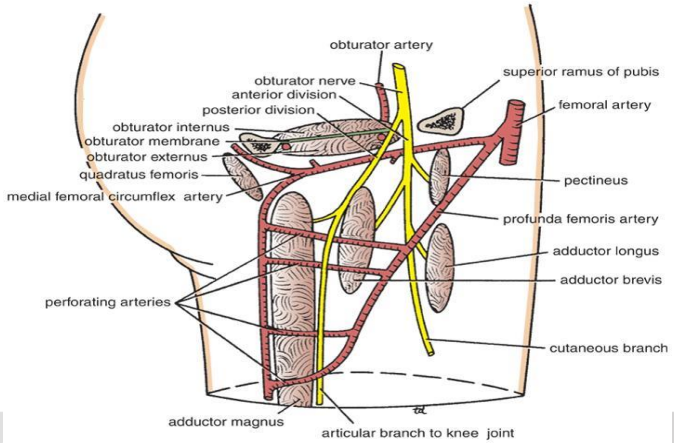


Obturator Nerve: Relations

- Emerge from the medial border of the psoas m.
- Cross the pelvic brim posterior to common iliac vessels
- Runs on the lateral wall of the pelvis between internal and external iliac vessels
- Accompanies the obturator vessels
- Exit through the obturator canal
- Divisons
  - Anterior
    - Anterior to obturator externus & adductor brevis mm.
  - Posterior
    - Traverse obturator externus m.
    - Posterior to adductor brevis m.
    - Anterior to adductor magnus m.



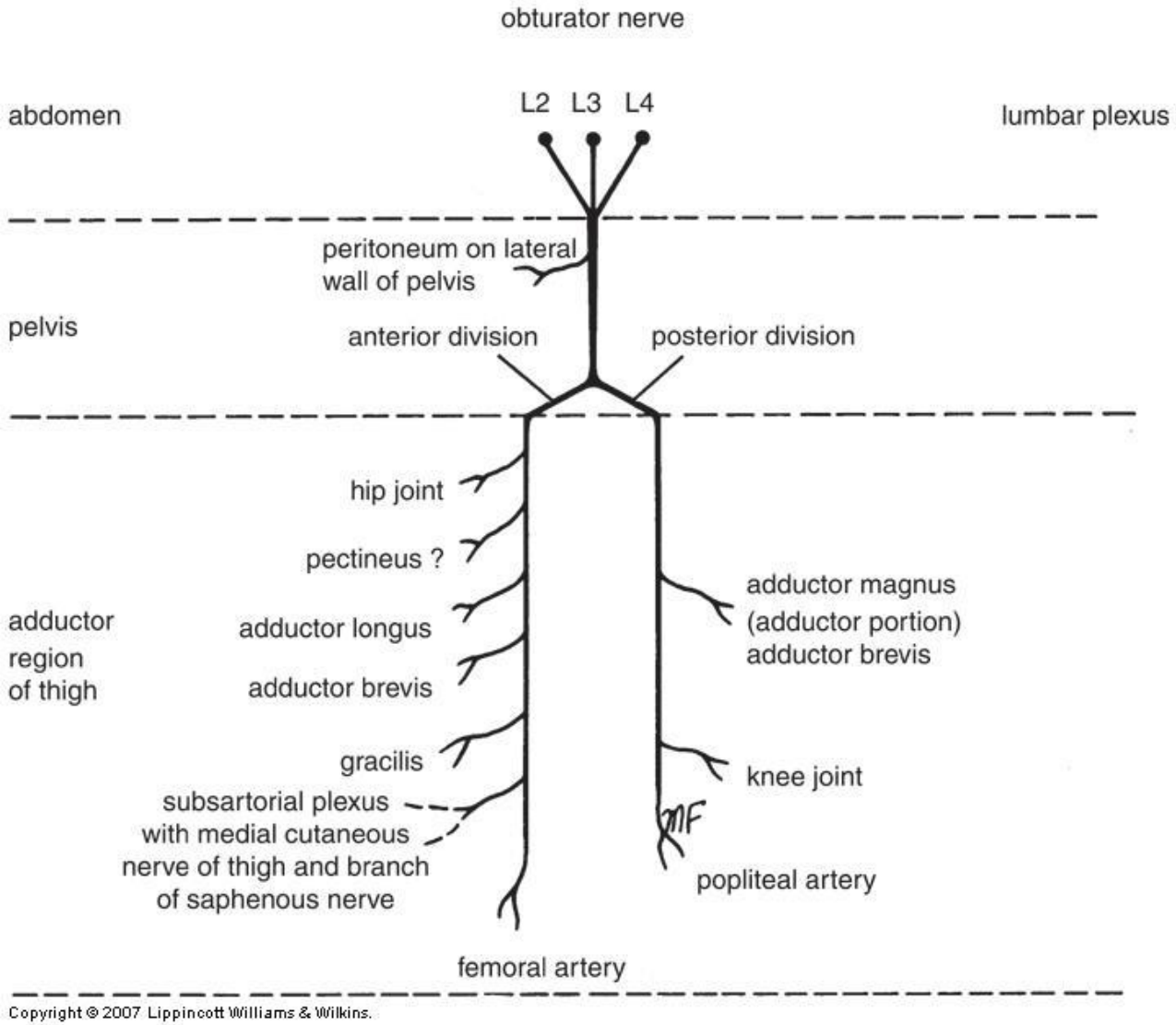
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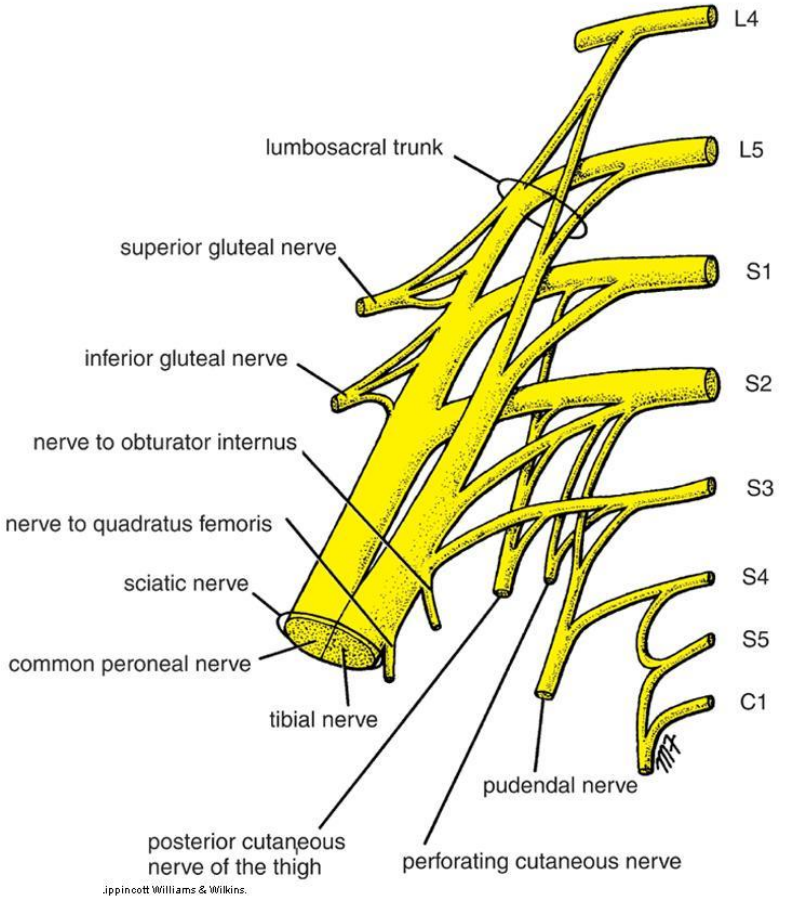
Obturator Nerve: Branches

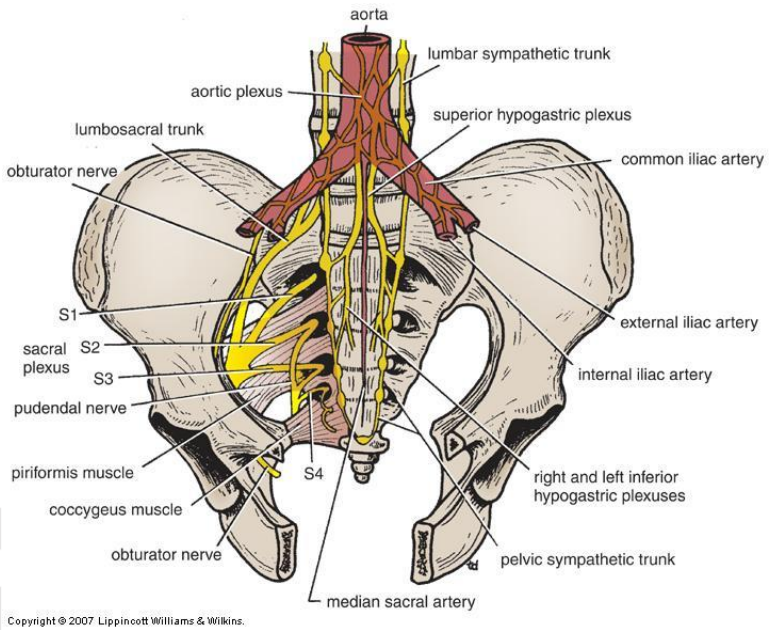
- Sensory n. to parietal peritoneum in pelvis
- Anterior division
  - Muscular branches
    - Gracilis, adductor brevis, adductor longus mm.
  - Cutaneous branch (medial side of the thigh)
  - Articular branch (hip)
- Posterior division
  - Muscular branches
    - Obturator externus & adductor magnus mm.
  - Articular branch (knee)



Sacral Plexus

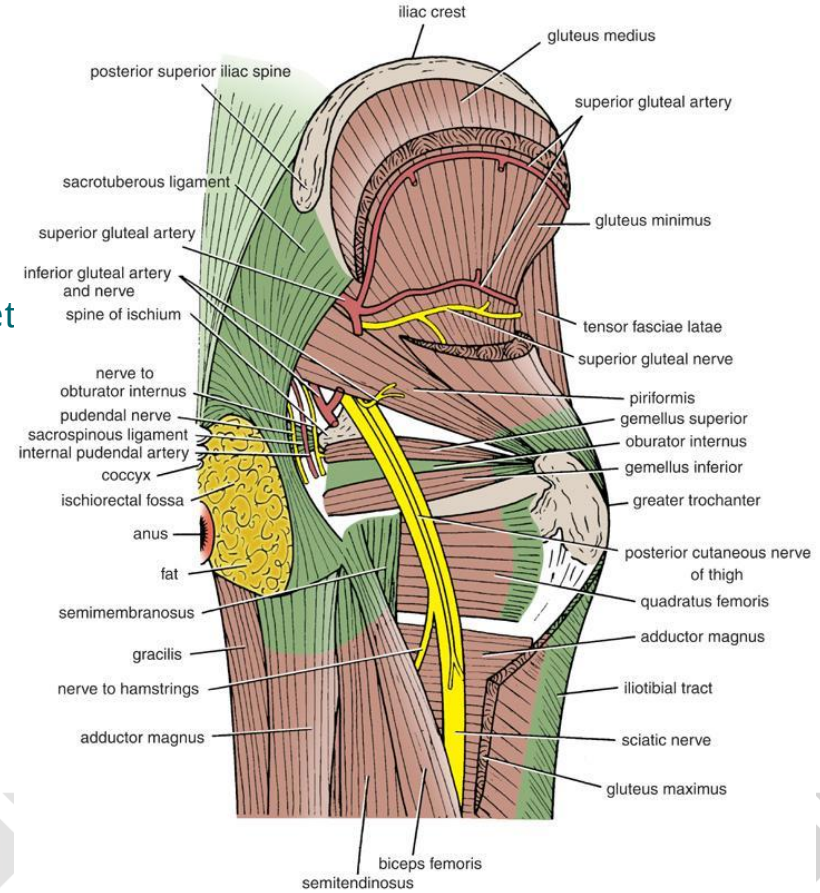
- Ventral rami L4-S4
- Anterior to the sacrum and posterior pelvic wall
- Form in front of periformis m.
- Supplies buttocks, perineum & part of lower limb
- Sciatic nerve = L4 to S3 supplies post. thigh & all below knee
- Branches exit from the greater sciatic notch inferior to periformis m. except the superior gluteal n.



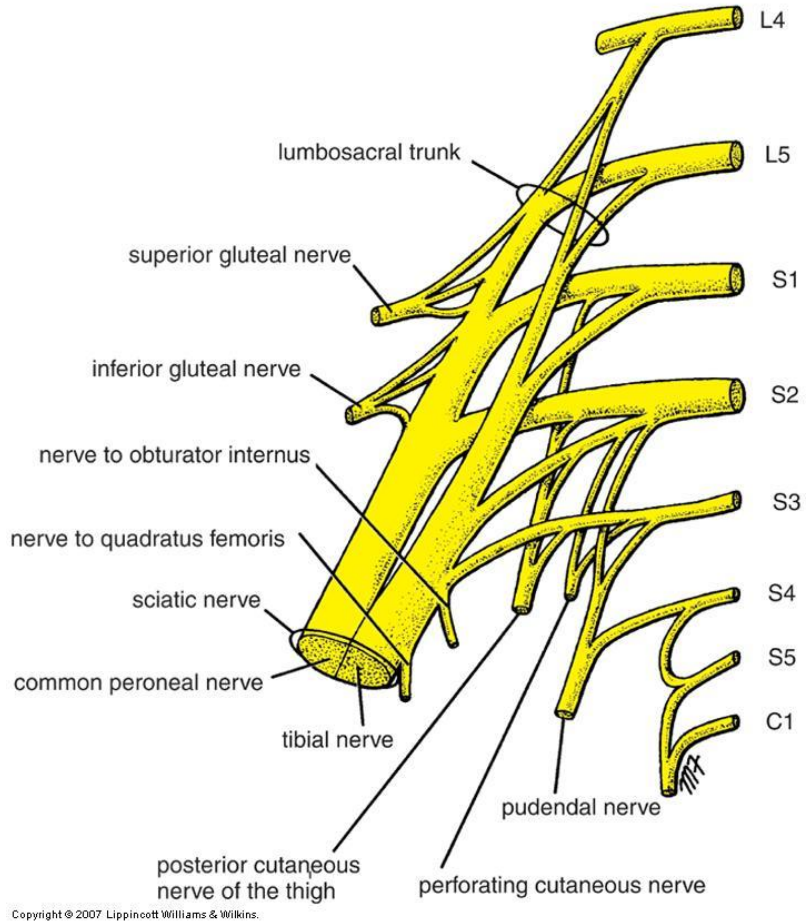


Sacral Plexus: Branches

- Sciatic nerve
  - Largest nerve in the body
- Superior gluteal n.
  - Gluteus medius and minimus and tensor fascia latae mm.
- Inferior gluteal n.
  - Gluteus maximus m.
- Nerve to quadratus femoris m.
  - Inferior gemellus m.
- Nerve to the obturator internus m.
  - ➢ Exit from greater sciatic notch and ret from the lesser sciatic notch
  - Superior gemellus m.
- Posterior cutaneous n. of the thigh
  - Buttock & back of the thigh
- Perforating cutaneous n.
  - Medial side of buttock
- Nerve to the piriformis m.
- Pudendal n. (perineum)



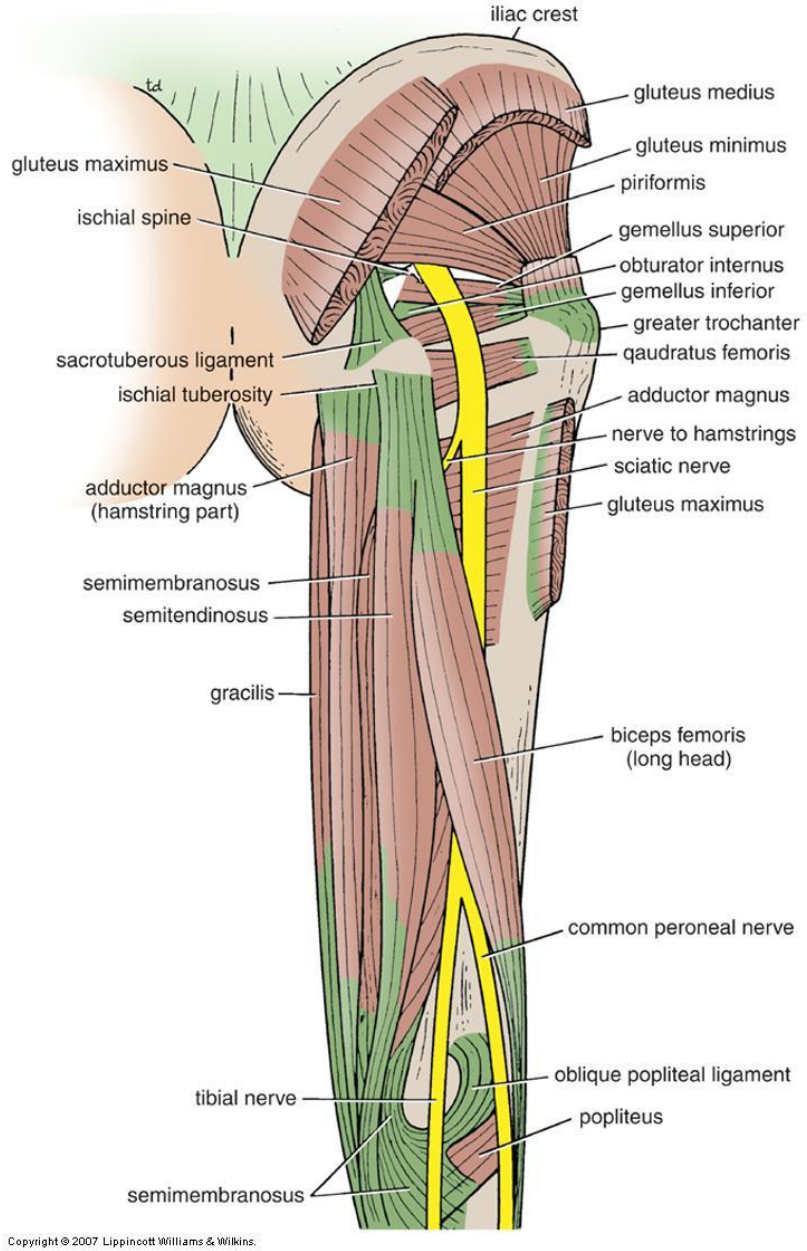
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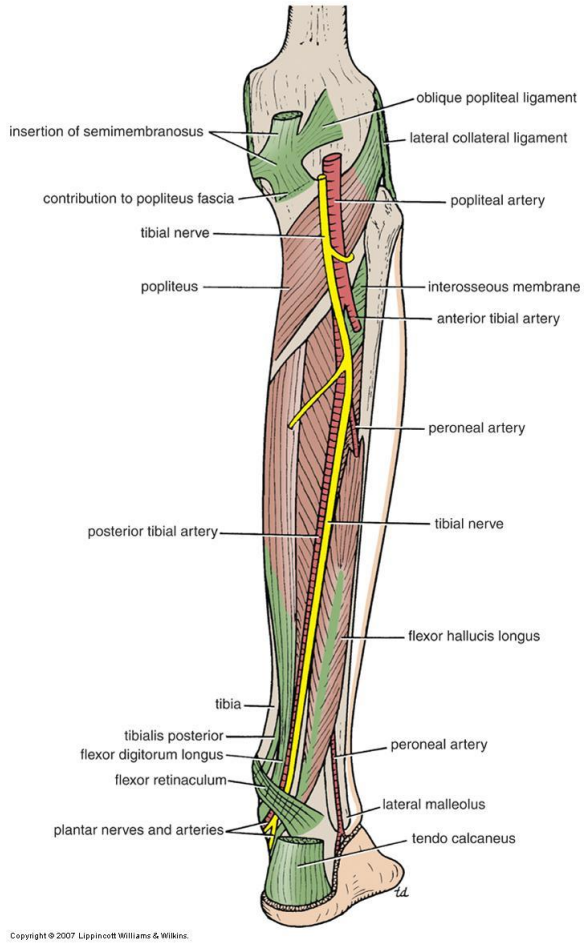
Sciatic Nerve: Relations

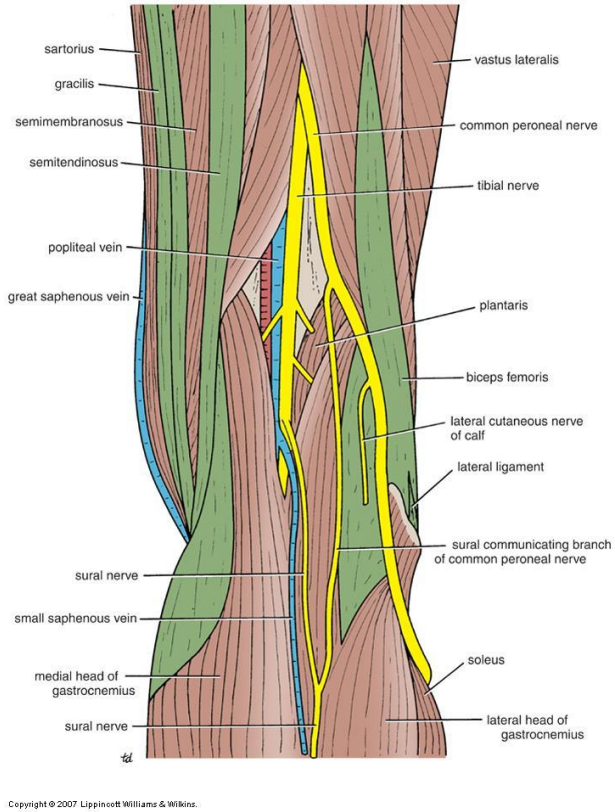
- Exit from greater sciatic notch
- Inferior to periformis m.
- Descend in the posterior thigh deep to gluteus maximus m. and biceps femoris m.
  - Supply the hamstring mm.
- At the superior part of the popliteal fossa divides into its terminal branches
  - Tibial n.
  - Common peroneal n.



Tibial Nerve

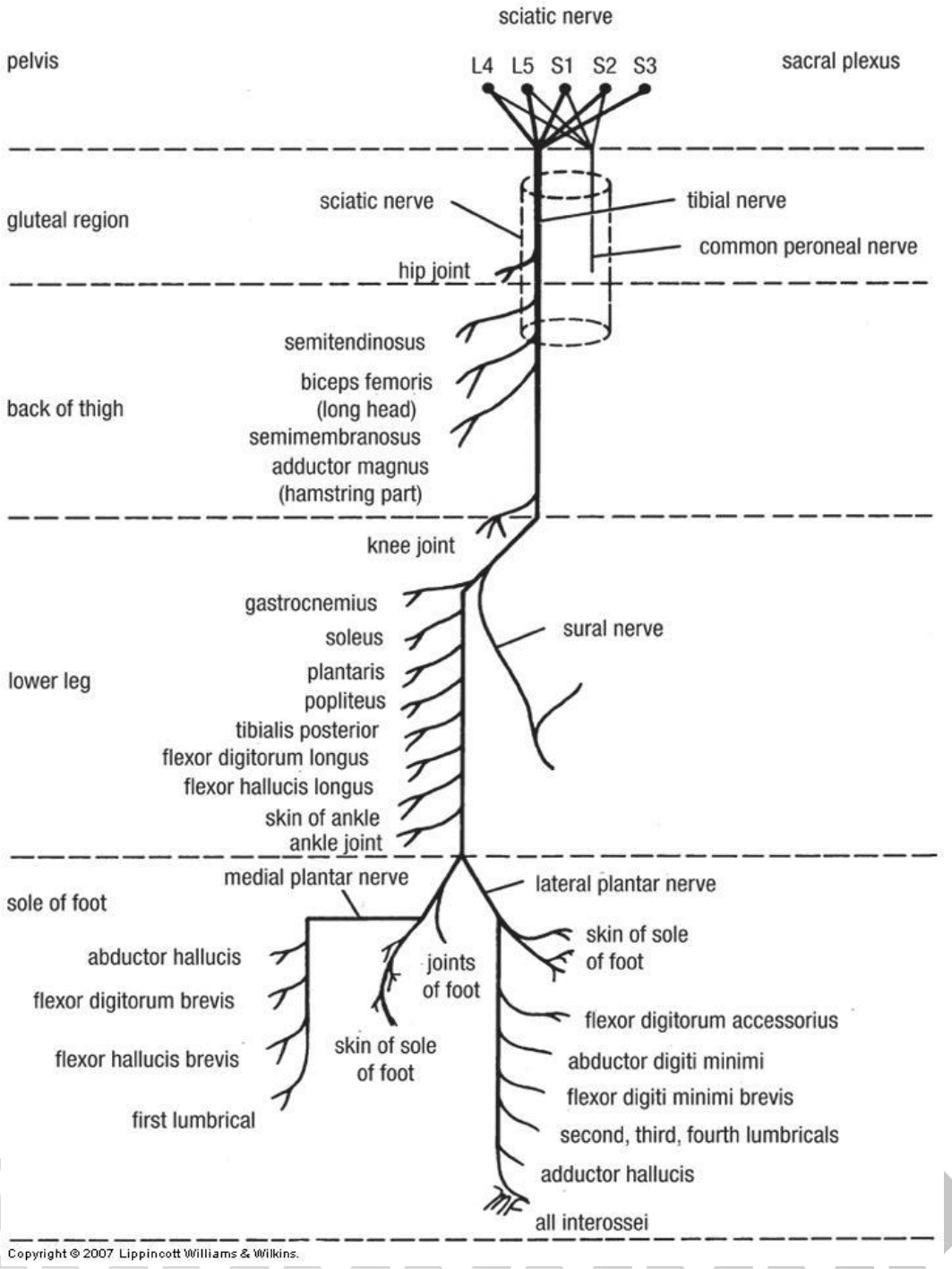
- Traverse the popliteal fossa
- Descend through the posterior compartment of the leg
- Deep to gastrocnemius and soleus mm.
- Deep to flexor retinaculum
- Divides into medial and lateral planter nn.





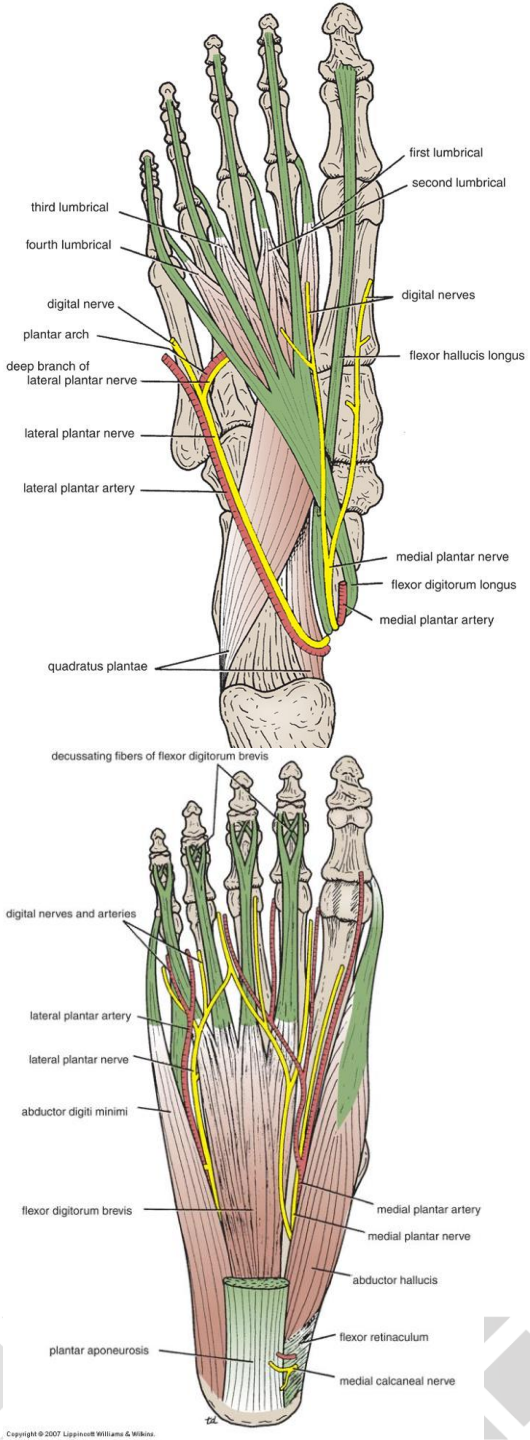
Tibial Nerve: Branches in Leg

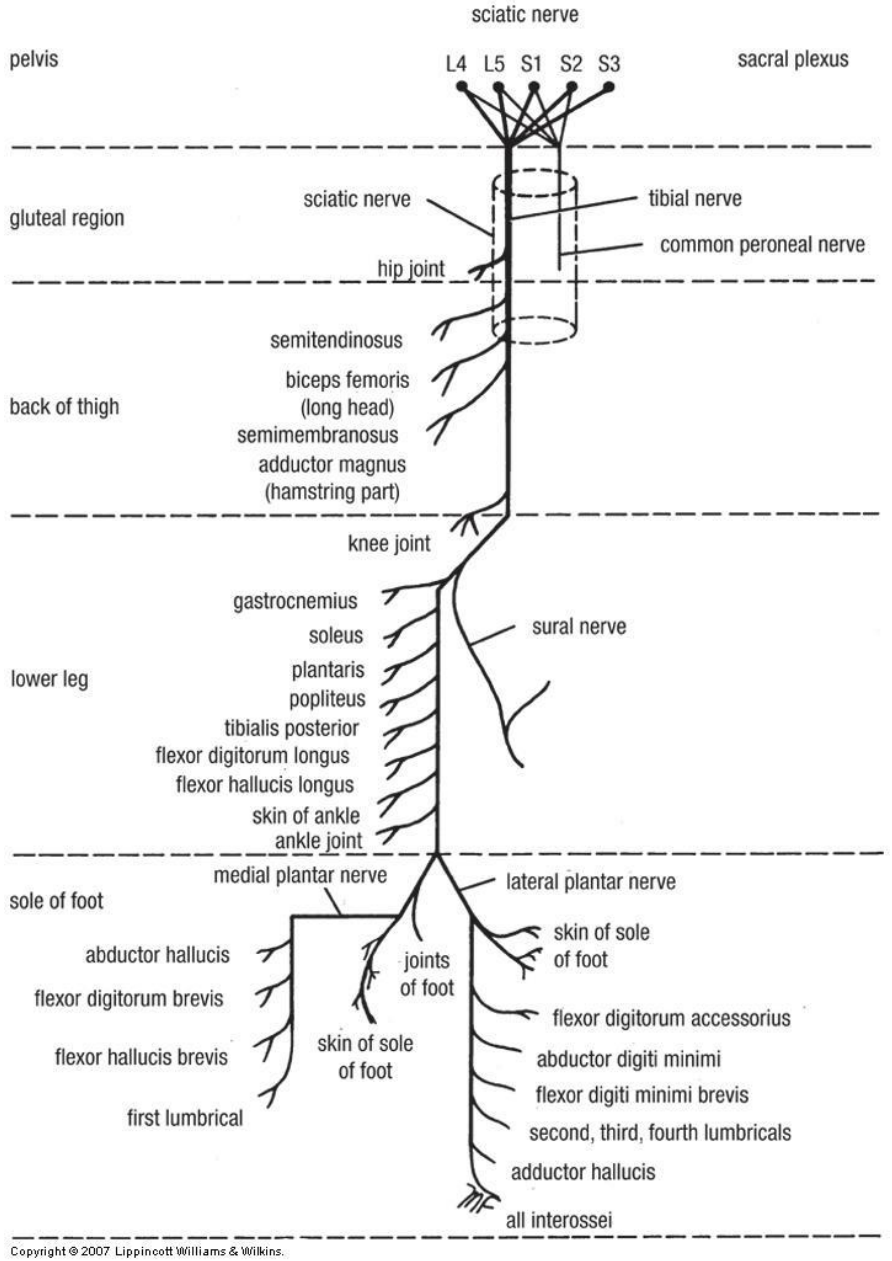
- Cutaneous branches
  - Sural n.
    - Back of the leg & lateral side of the foot
  - Medial calcaneal n.
    - Skin over medial side of heel
- Muscular branches
  - Muscles of the posterior compartment of the leg
- Articular branches (knee & ankle)



Tibial Nerve: Branches in sole

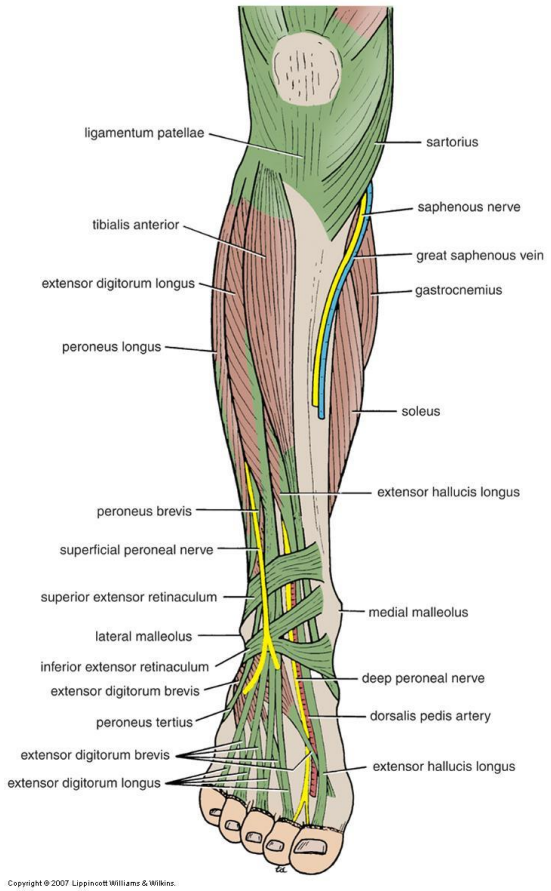
- Medial planter n.
  - ➢ Companies medial planter a.
  - ➢ Deep to abductor hallucis m.
  - Cutaneous branch
    - Medial side of sole & 3½ toes
  - Muscular branch
    - Abductor hallucis, flexor digitorum brevis, flexor hallucis brevis, & 1st lumbrical mm.
- Lateral planter n.
  - ➢ Companies lateral planter a.
  - ➢ Deep to abductor hallucis & flexor digitorum brevis mm.
  - Cutaneous branch
    - Lateral side of sole & 1½ toes
  - Muscular branch
    - Abductor digiti minimi, flexor digiti minimi brevis, adductor hallucis, 2-4 lumbricals, & interosseous mm.

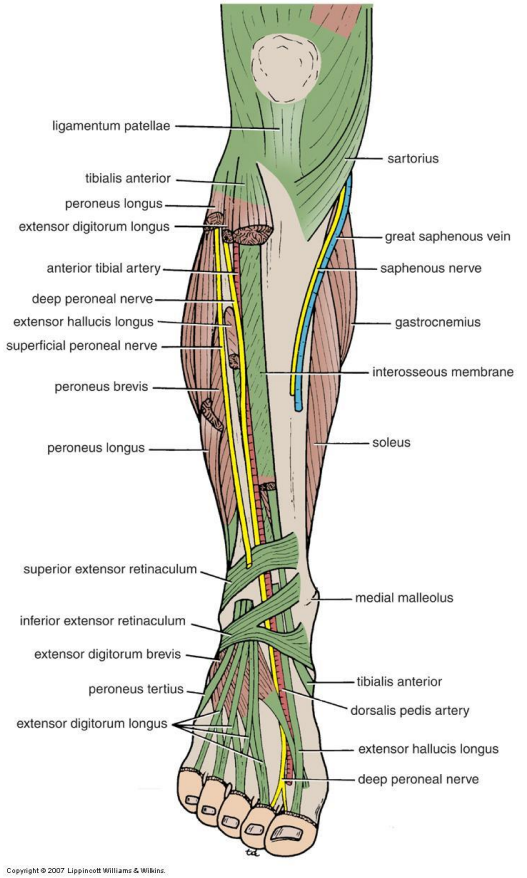




Common Peroneal Nerve

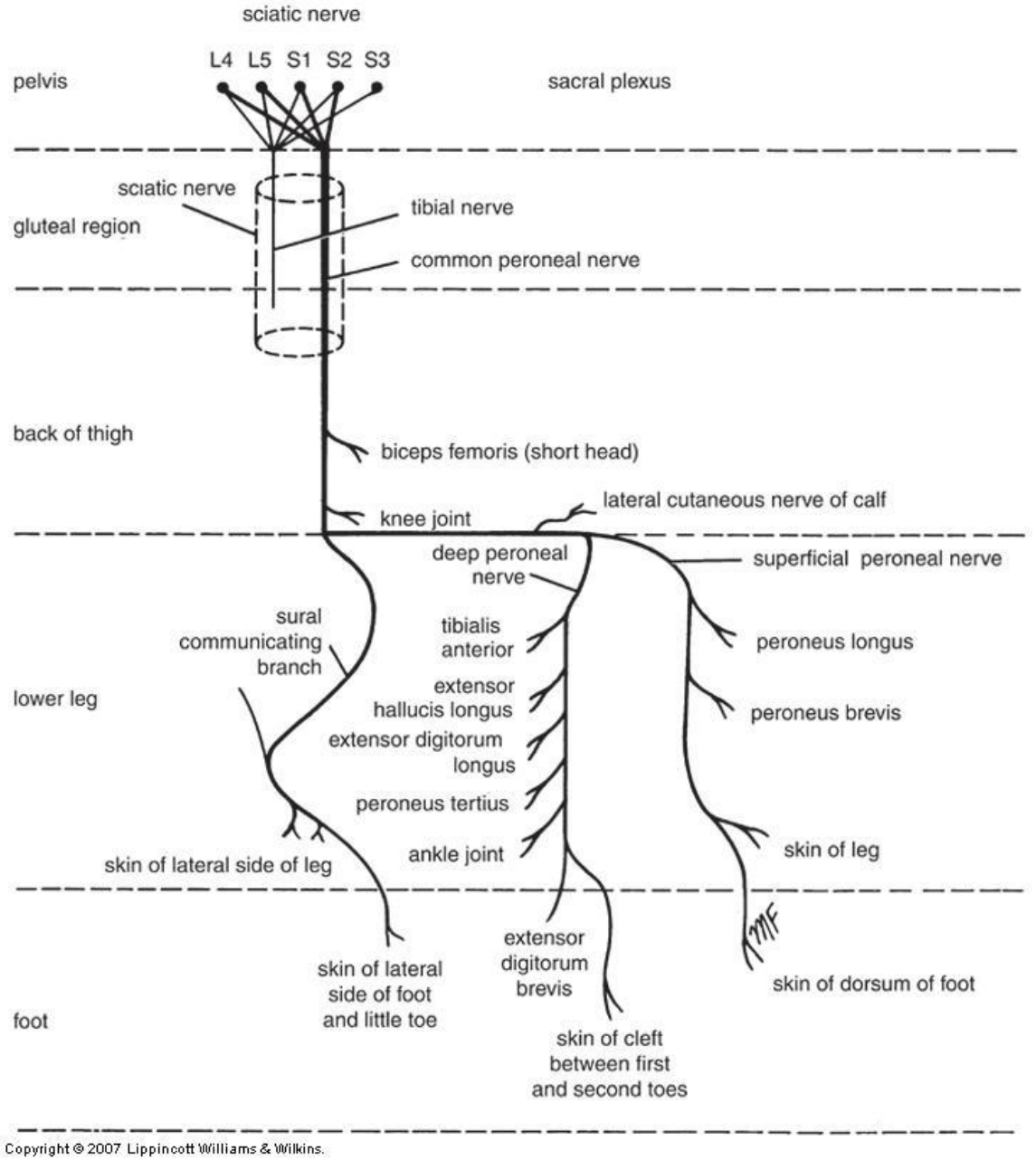
- Traverse the popliteal fossa
- Around head of fibula
- Traverse the peroneus longus m.
- Divide into terminal branches
  - Superficial peroneal n.
    - Descends in the lateral compartment between peroneus longus & previs mm.
  - Deep peroneal n.
    - Descends in the anterior compartment deep to the extensor digitorum longus m.
    - Anterior to the interosseous membrane
    - Accompanies the anterior tibial vessels





Common Peroneal Nerve: Branches

- Cutaneous branches
  - Sural communicating branch
  - Lateral cutaneous n. of the calf
- Muscular branch
  - Short head of biceps femoris m.
- Articular branch (knee)
- Superficial peroneal n.
  - Cutaneous branch
    - Skin over the lower anterior leg and dorsum of foot
  - Muscular branch
    - Lateral compartment
- Deep peroneal n.
  - Cutaneous branch
    - Between 1st & 2nd toes
  - Muscular branch
    - Anterior compartment & extensor digitorum brevis m.
  - Articular branch (ankle & tarsal)



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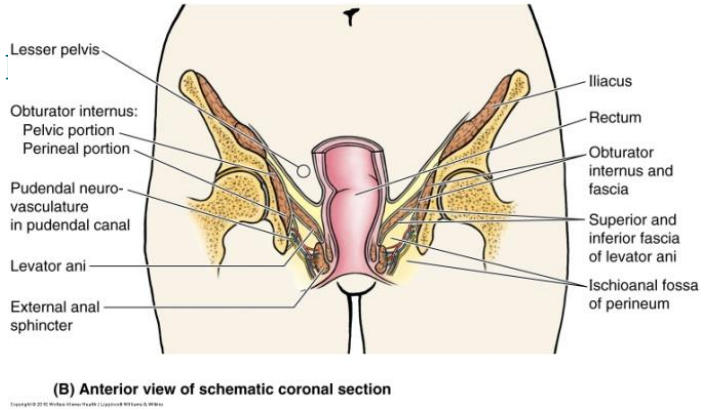
Pudendal Nerve

- Exit from greater sciatic notch and return from the lesser sciatic notch
- Traverse the pudendal canal with the internal pudendal vessels in the ischiorectal fossa

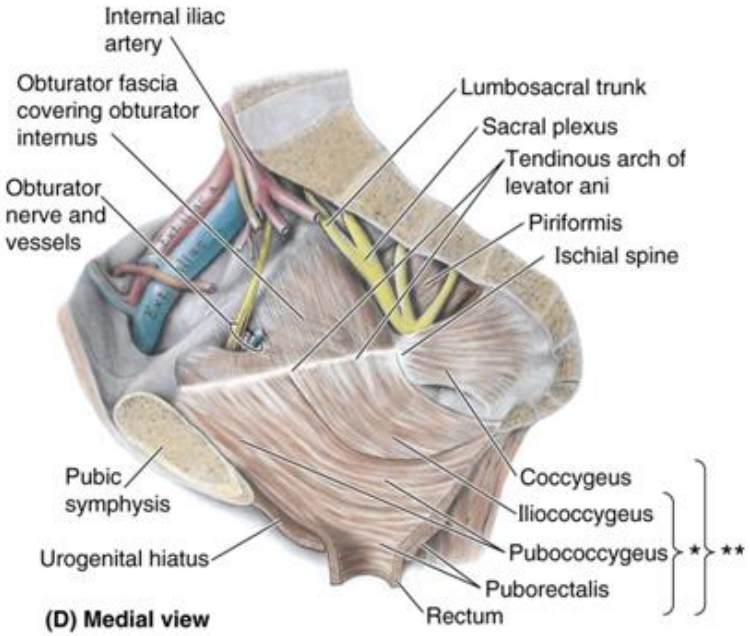
❑Branches

- Inferior rectal n.
  - Sensory
    - Mucus membrane of lower half of anal canal & skin
  - Muscular
    - External anal sphincter
- Perineal n.
  - Cutaneous (posterior scrotal (labial) n.)
    - Posterior surface of scrotum or labia majora
  - Muscular
    - Superficial and deep transverse perineal mm., bulbospongiosus, ischiocavernosus mm., external urethral sphincter, and levetor ani mm.
- Dorsal n. of the penis (clitoris)

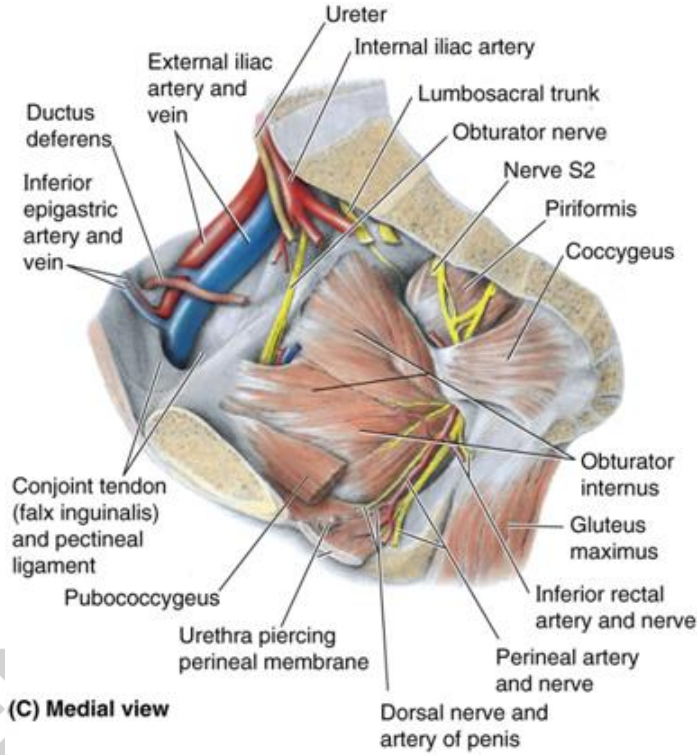
Skin and deeper structures of the penis (clitoris) •



(B) Anterior view of schematic coronal section



(D) Medial view

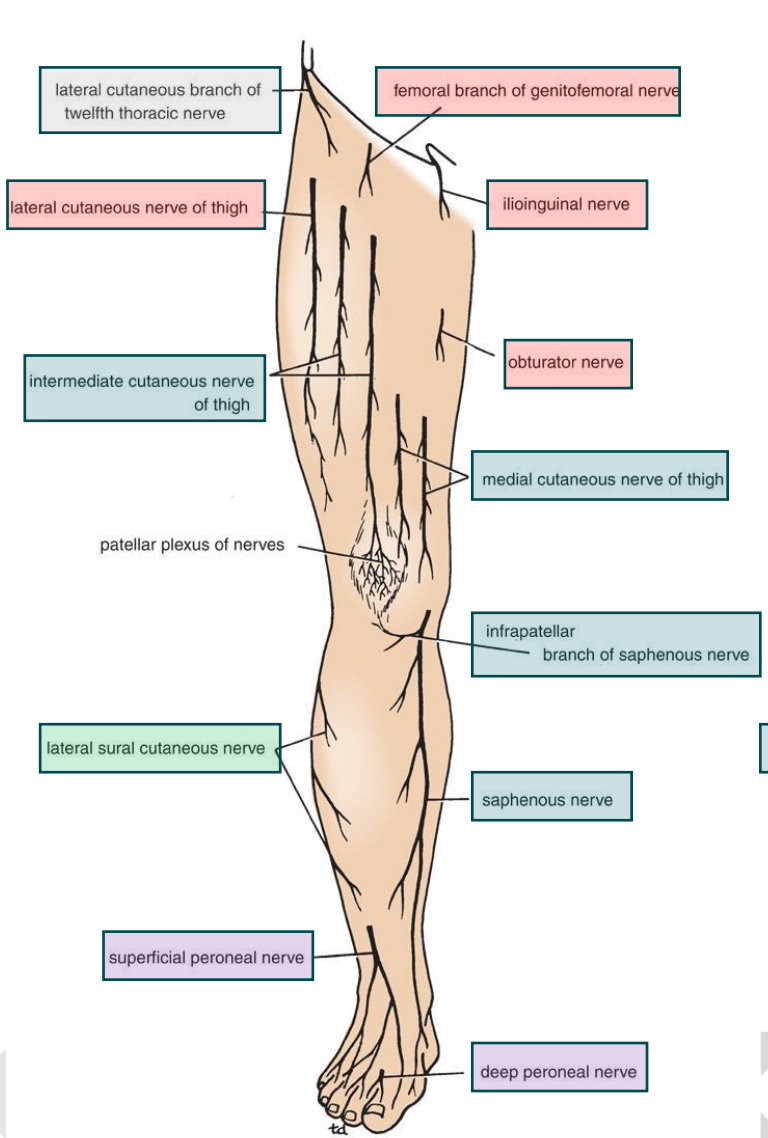


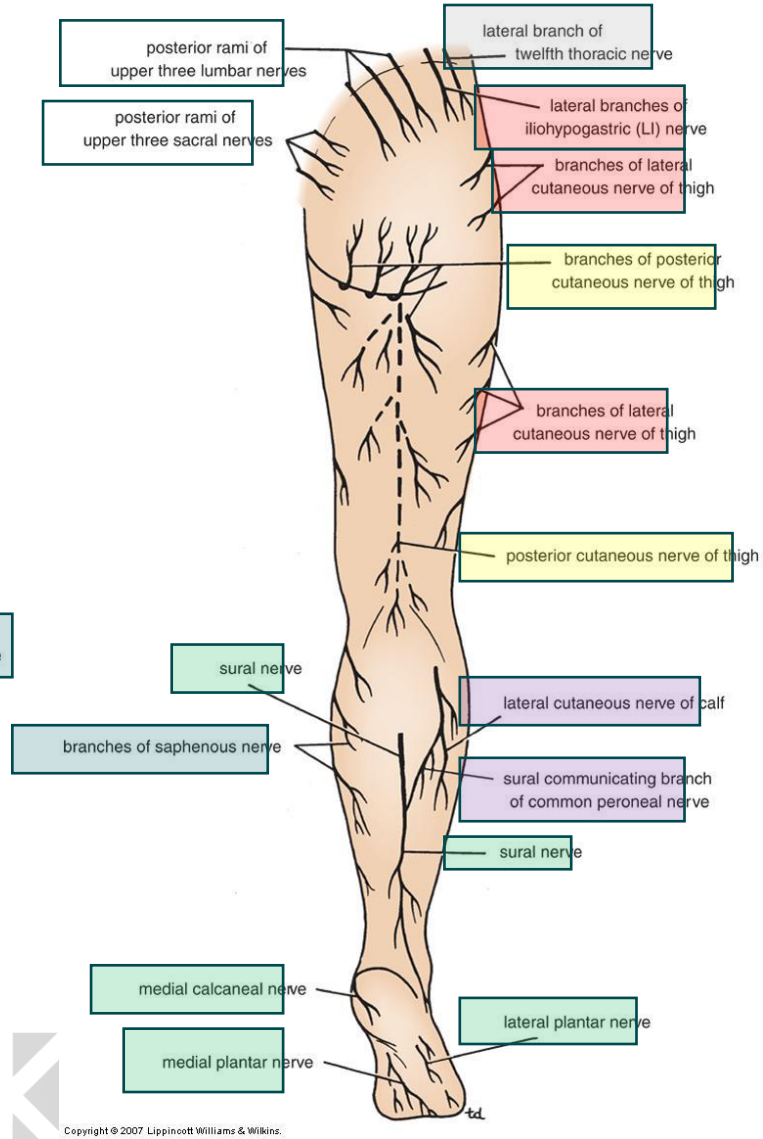
(C) Medial view

Cutaneous Innervation of the Lower Limb

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- Posterior rami
- 12th thoracic
- Lumber plexus
- Femoral
- Sacral plexus
- Tibial
- Common peroneal







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Lower Limb: Peripheral Nerve Lesions

Femoral Nerve Injury

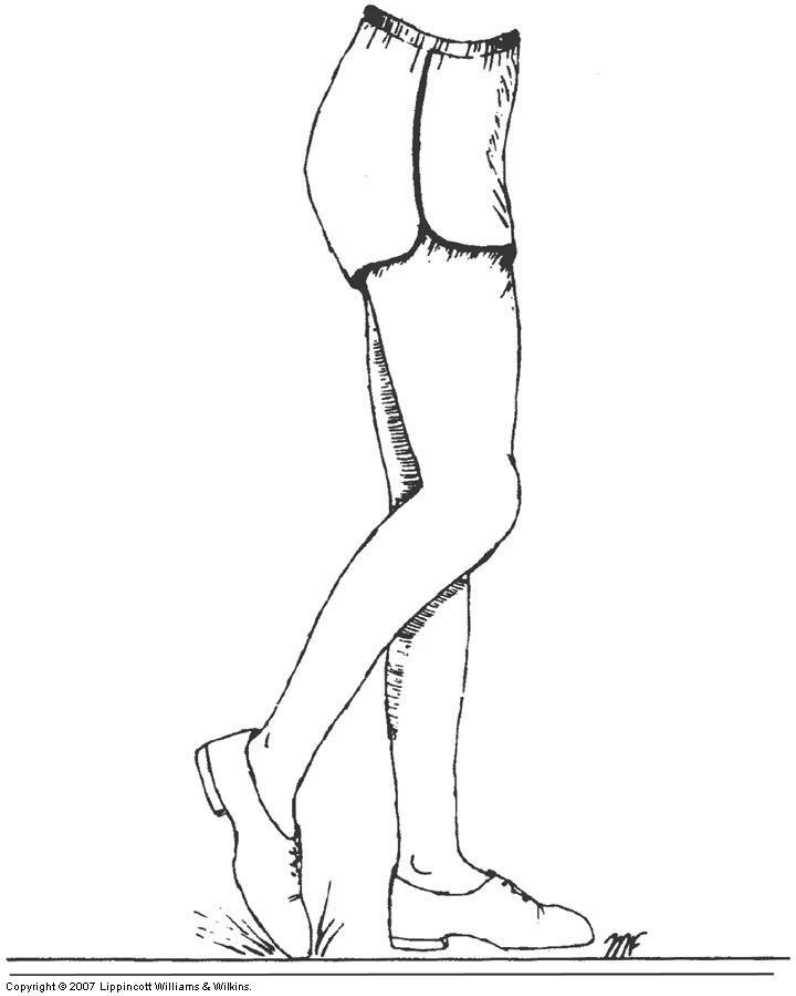
- Results from
  - Stab or gunshot wound
  - Complete division is rare
- Paralysis of quadriceps femoris m.
  - Knee can not be extended against resistance
  - Patient usually press against the distal thigh during walking
- Loss of sensation along the medial part of the lower limb and the anterior part of the thigh

Obturator Nerve Injury

- Rare
- Paralysis of the adductor muscles
- Loss of sensation of small area of the medial part of thigh

Sciatic Nerve Injury

- Results from
  - Penetrating wounds, fractures of the pelvis, or dislocation of the hip bone
  - Faulty IM injections in the gluteal region
- Complete injury is rare
  - 90% of the cases affect the common peroneal part (more superficial)
- Paralysis of the hamstring muscles and all muscles below knee
  - Foot droop (planter flexed position)
- Loss of sensation below knee except for the medial part (femoral n.)
  - Trophic ulcers of the sole

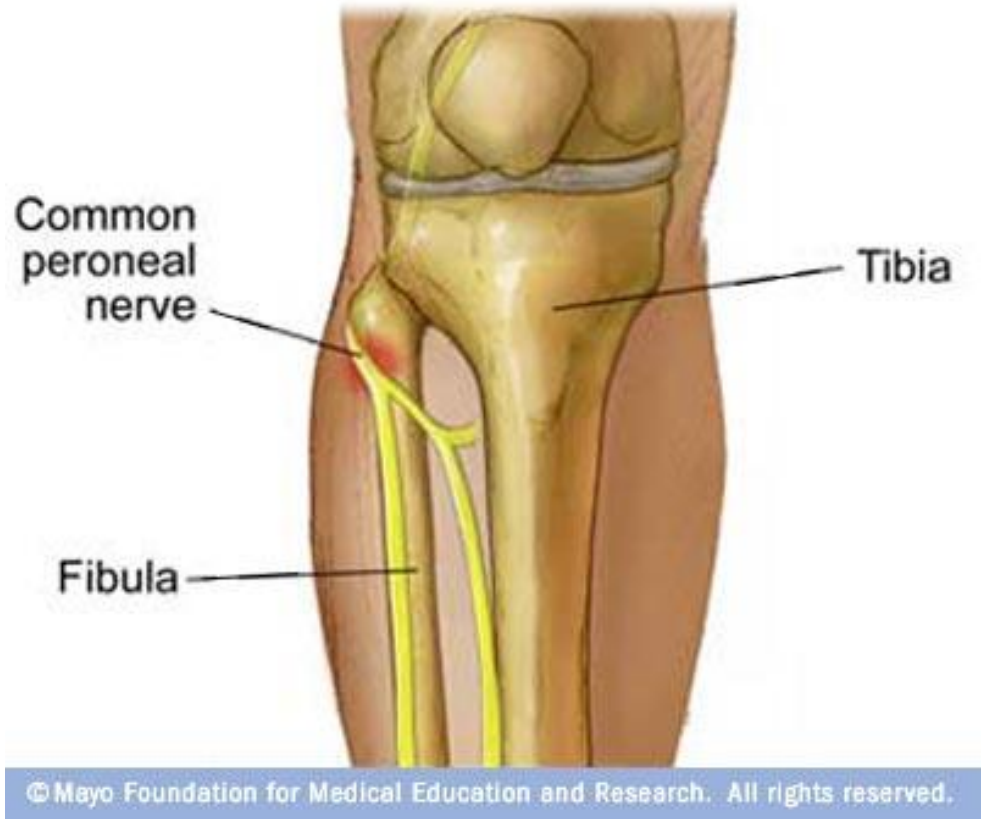

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Sciatica

- Pain along the sensory distribution of the sciatic nerve
  - Posterior of thigh
  - Posterior & lateral sides of leg
  - Lateral part of foot
- Results from
  - Prolapse of an IVD (pressing on the roots of the spinal nerves)
  - Pressure on the sacral plexus or sciatic n. by tumor

Common Peroneal Nerve Injury

- Results from
  - Fractures of the neck of the fibula
- Paralysis of the muscles of the anterior and lateral compartments of the leg
  - Equinovarus (Planter flexion (foot drop)and inversion)
- Loss of sensation on the anterior and lateral sides of leg & dorsum of foot



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Tibial Nerve Injury

- Rare
- Paralysis of the muscles in the posterior compartment of the leg and the muscles of the sole
  - Calcaneovalgus (Dorsiflexion & eversion of foot)
- Loss of sensation on the sole of the foot
  - Trophic ulcers

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